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Attenuation correction factors for cylindrical, disc and box geometry

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ABSTRACT

In the present study, attenuation correction factors have been experimentally determined for samples having cylindrical, disc and box geometry and compared with the attenuation correction factors calculated by Hybrid Monte Carlo (HMC) method [C. Agarwal, S. Poi, A. Goswami, M. Gathibandhe, R.A. Agrawal, Nucl. Instr. and. Meth. A 597 (2008) 198] and with the near-field and far-field formulations available in literature. It has been observed that the near-field formulae, although said to be applicable at close sample-detector geometry, does not work at very close sample-detector configuration. The advantage of the HMC method is that it is found to be valid for all sample-detector geometries.

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1. Introduction

The Non-destructive assay (NDA) technique is important for assaying nuclear materials which may have varied shapes and sizes. Assay of such samples by gamma ray spectrometry suffers from absorption of gamma rays within the sample. The attenuation of gamma rays may be due to the high Z of the element monitored as in the case of nuclear materials where U, Pu, Th are present in bulk amount and cause the absorption of their gamma rays themselves. Apart from nuclear materials, absorption of gamma rays within the sample may also result due to voluminous matrix as may be the case in environmental samples where generally trace amounts of U and Th are present and large sample may be required for counting. In both the circumstances, the proportionality between the measured count rate and the mass of the nuclide monitored is lost due to self-attenuation. In such cases, the use of standards can solve the problem only when they have equivalent geometry and density as the sample to be analyzed. Due to the diverse nature of the samples encountered in NDA, mostly this condition is not fulfilled. Determination of attenuation correction factor is therefore necessary to increase the linear dynamic range of concentration that can be determined by gamma ray spectrometry for the required sample-detector geometry. Appropriate attenuation correction also makes the measurement independent of the sample matrix. The corrected count rate (CR) of an attenuating sample is given by

$$CR = k_{att} \times k_{rate} \times RR \tag{1}$$

where RR refers to the raw count rate of the sample and k_{att}, k_{rate} are the correction factors for attenuation in the sample and for rate-related losses which includes electronic loss as well as summing effect, respectively. There are several approaches available in literature for attenuation correction [1–13]. General attenuation correction procedures have been explained in detail in [14].

A Hybrid Monte Carlo (HMC) method developed for attenuation correction has been discussed in our previous paper [15]. This is a simple numerical method in which random points are generated simultaneously on the detector surface and within the sample volume. The line connecting these two points represents the path followed by the gamma ray to reach the detector. Two such representative gamma ray paths are shown in Fig. 1 for a cylindrical sample whose axis is perpendicular to the detector axis. The centre of the cylinder is considered to be the origin. 'd' is the sample-to-detector distance, from the origin to the centre of the circular detector surface. If the distance traveled by the gamma ray within the sample is denoted by t, then k_{att} can be expressed as

$$k_{att} = \frac{\langle 1/r^2 \rangle}{\langle \exp(-\mu_L t)/r^2 \rangle} \tag{2}$$

where r is the distance traveled by the gamma ray to reach the detector surface and μ_L is the linear attenuation coefficient (cm^{-1}) for the gamma ray of interest. The numerator represents the average probability of a gamma ray from an unattenuated sample reaching the detector. The exponential term in the denominator

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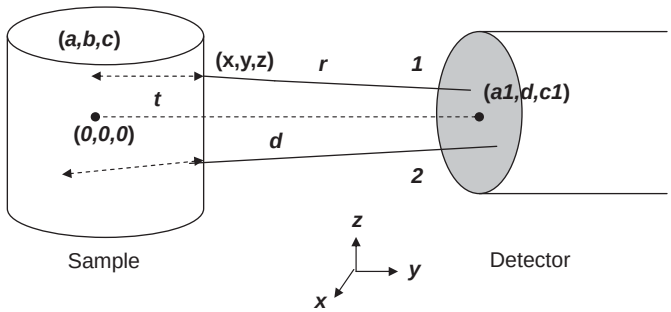

Fig. 1. Representative paths of two gamma rays from a cylindrical sample to the detector surface.

accounts for the attenuation correction within the sample. For each pair of random points on the sample and the detector, t is an unknown which is obtained by calculating the co-ordinate of the point of intersection of the sample surface and the straight line joining the two random points. The calculation requires linear attenuation coefficient for the gamma ray energies monitored as an input. The sample dimensions like its radius and height and detector radius is also given as input. The methodology has been explained in detail in [15]. The approach has been applied to some of the common geometries like cylindrical, spherical and box-shaped samples. The computed k_{att} values for all the three geometries were found to compare well with the k_{att} values calculated using the expressions available in [14] and from Monte Carlo simulation over a wide range of transmittance. It was observed in the previous paper that the HMC method avoids any simplified assumption generally used in the near- or far-field geometry formula. Moreover, unlike the far-field and near-field approaches, the detector is not considered to be a point detector. This method does not require detailed information about detector dimensions and only the radius of the detector crystal (considered to be a cylinder) comes into picture in the calculations.

In order to validate the attenuation correction procedure developed in ref. 15, attenuation correction factors have been measured for samples of different geometry, and compared with the results of HMC calculations. The experiment was performed for cylindrical, disc and box geometry. The k_{att} data has also been compared with the k_{att} values determined from Monte Carlo simulation for cylindrical and disc geometries. Applicability of different analytical formulations available in the literature is discussed in the light of present experimental results and HMC calculations.

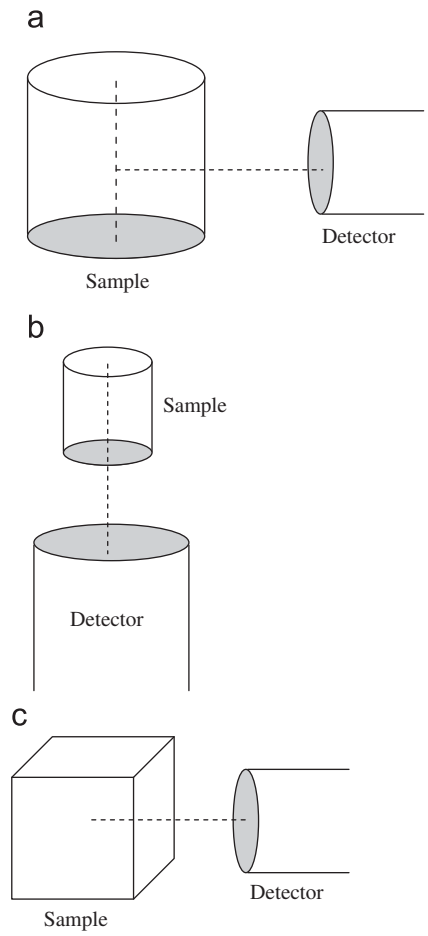

Fig. 2. The geometrical arrangement of the sample and the detector: (a) cylindrical geometry, (b) disc geometry and (c) box geometry.

Table 1
Gamma ray energies of ^{169}Yb and their corresponding abundances.

Gamma ray energy (keV)	Abundance (%)
130.5	11.3
177.2	22.2
198.0	35.8
307.7	10.0

2. Experimental

Attenuation correction factors were determined experimentally for cylindrical, disc and box geometry. For cylindrical and disc geometry, uranium solution was chosen as the attenuating matrix. For this purpose, four 20 ml standard uranyl nitrate solutions in the concentration range of 100–390 mg/ml were prepared by dissolving the required amount of U_3O_8 in 2 M HNO_3 in standard cylindrical vials. The cylindrical and disc geometry were manifested by keeping the cylindrical sample such that its axis is perpendicular to the detector axis (cylindrical geometry) and coaxial with the detector axis (disc geometry). Both the geometrical set-ups are shown in Fig. 2(a) and (b). These geometries were realized by counting the same set of uranyl nitrate samples in a horizontal and a vertical HPGe detector, respectively. Both the detectors had a relative efficiency of 20% and a resolution of 2.2 keV at 1.33 MeV. The 143 and 186 keV gamma rays emitted by ^{235}U were monitored. The efficiency of the required geometry was obtained using 20 ml ^{152}Eu and ^{133}Ba standard solutions. The μ_L of the uranium solutions, which is required as an input for the HMC approach, was measured experimentally for all the four samples at the energies of interest, by transmission measurement of a collimated beam of gamma ray. For this, the transmission source was counted with and without the samples. The ^{169}Yb was used as the transmission source as it has gamma rays close to the ^{235}U gamma rays used in our measurement. Table 1 gives the ^{169}Yb gamma ray energies and their corresponding abundances used for transmission measurement. Here, the unattenuated count rate was obtained by replacing the sample with 2 M HNO_3 solution. The transmission curve for a representative uranyl nitrate solution as a function of gamma ray energy is shown in Fig. 3. The transmittance for ^{235}U energies was obtained by linear interpolation between the transmittance for ^{169}Yb energies. The symbols (+) in the figure shows the gamma ray energies of ^{235}U monitored. The linear attenuation coefficient was obtained from

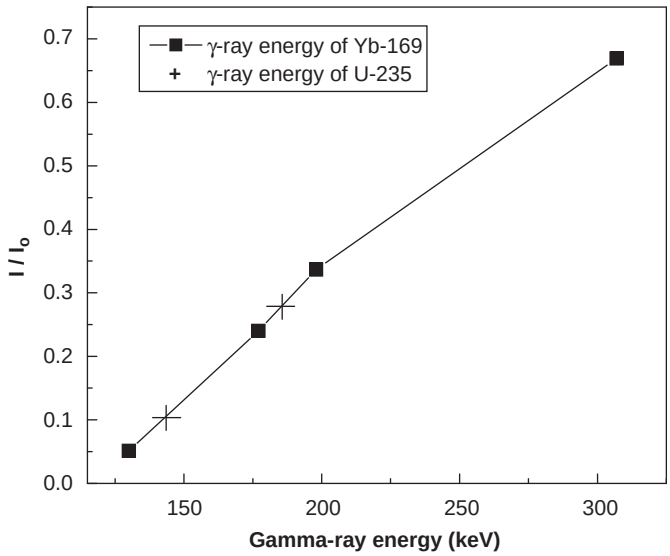

Fig. 3. Transmission curve of a uranium solution as a function of gamma ray energy using ^{169}Yb as a transmission source.

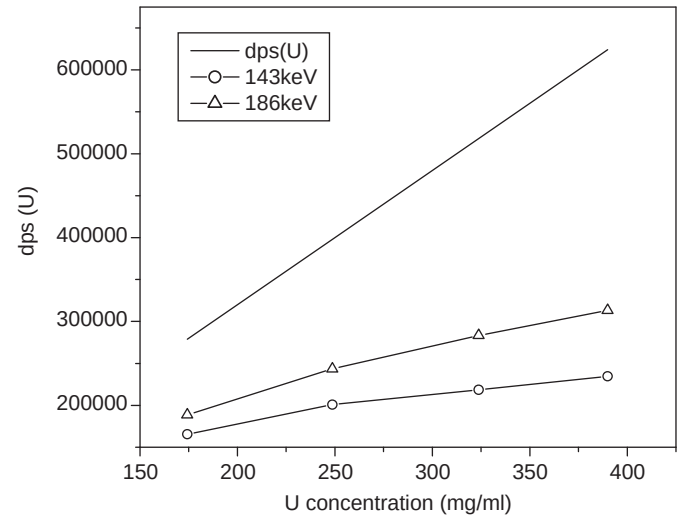

Fig. 4. Experimental disintegration rate of uranium at different energies for disc geometry as a function of uranium concentration. The straight line shows the expected dps from the samples as a function of uranium concentration.

the relation

$$\mu_L = -\ln(I/I_o)/x \tag{3}$$

where I and I_o are the intensities of gamma ray with and without the sample, respectively, x is the distance traversed by the gamma ray through the sample, i.e. the sample diameter in the case of cylindrical samples.

For the box-shaped samples, lead acetate solutions were used as the attenuating matrix, keeping in mind the high Z of lead and high solubility of lead acetate. These solutions in the concentration range of 10–400 mg/ml were prepared by dissolving the required amount of lead acetate in water. 40 ml of the solutions were taken in cubic plastic vials. The linear attenuation coefficients of these samples were measured by using ^{57}Co as the transmittance source by the method described above. To get experimental attenuation correction factors, known amount (dps/g) of ^{57}Co activity in solution form was added to the samples. Known amount of activity was also added to a water sample which was chosen as a blank. These samples along with the blank were then counted in a 20% HPGe detector having a resolution of 2.2 keV at 1.33 MeV. The 122 keV gamma ray of ^{57}Co was monitored. The experimental set-up for the box geometry is shown in Fig. 2(c).

3. Results and discussion

3.1. Cylindrical and disc geometry

Fig. 4 shows the experimental disintegration rate for disc geometry corresponding to different energies as a function of uranium concentration in the sample. These were calculated by using the standard expression

$$dps_{\exp t} = \frac{CR_\gamma}{\varepsilon_\gamma a_\gamma} \tag{4}$$

where, CR_γ, ε_γ, a_γ refers to the count rate from sample, detection efficiency and abundance of the gamma ray concerned, respectively. The straight line in the figure is the dps of the U-solution (dps_{calc}) calculated from the known uranium concentration. It can be seen from the figure that the experimental dps deviates from linearity. This deviation increases with the U-concentration and with the decreasing gamma ray energy. This points to the effect of gamma ray attenuation by the U-solutions. The experimental attenuation correction factors ($k_{att(\exp)}$) were obtained from the experimental dps as follows:

$$k_{att(\exp)} = dps_{calc}/dps_{\exp t} \tag{5}$$

Measurements were also carried out at different sample-to-detector distances so as to check the applicability of the HMC approach over a wide range of sample-to-detector distances.

To calculate the attenuation correction factors using the HMC method, the dimensions of the real sample were given as an input. The diameter of the cylindrical detector used in the calculations was 5 cm for horizontal detector (cylindrical geometry) and 6 cm for vertical detector (disc geometry) as obtained from their respective manuals. Along with this, the linear attenuation coefficients obtained experimentally at the energies monitored for different samples were given as an input in the respective FORTRAN programs for cylindrical and disc geometry. The dimensions of the samples, the corresponding measured densities and the linear attenuation coefficients are given in Table 2. Separate runs were taken for each sample-to-detector distance. All calculations were performed by generating 10^5 particles, using a desktop PC. The time taken for each computation was less than 5 min in all the cases.

In this study, Monte Carlo simulation for both the geometries was also performed using MCNP code [16]. In these calculations, apart from the precise information about the sample-detector geometry, the concentration of uranium in the sample was given as an input. The unattenuated count rate was obtained by replacing the sample solution with air. The k_{att} from MCNP was calculated from the ratio of unattenuated to attenuated count rates. Fig. 5 shows the theoretical k_{att} values calculated from the HMC method compared with the experimental and MCNP results for cylindrical geometry at $d = 2.5$, 4.5 and 20 cm as a function of transmittance. These k_{att}'s have also been compared with the near-field k_{att} so as to see whether the HMC method converges with the prediction of the near-field formula (two-dimensional model) [14] for cylindrical geometry

$$k_{att} \frac{\sum_{m=1}^{M}\sum_{n=1}^{N} \Delta A(m,n)/L^2(m,n)}{\sum_{m=1}^{M}\sum_{n=1}^{N} \{\exp[-\mu t(m,n)]\}\Delta A(m,n)/L^2(m,n)} \tag{6}$$

Table 2
The details of sample dimensions, its density, gamma-ray energies monitored and corresponding linear attenuation coefficients.

Sample no.	Cylinder (Inner dia: 2.54 cm, Height: 4 cm)			Box ($3.6 \times 3.6 \times 3.6\,cm^3$)		
	Density (g/cc)	Gamma-ray energy (keV)	μ_L (cm^{-1})	Density (g/cc)	Gamma-ray energy (keV)	μ_L (cm^{-1})
1.	1.593	143	1.02	1.254	122	0.56
		186	0.59			
2.	1.503	143	0.84	1.209	122	0.44
		186	0.49			
3.	1.421	143	0.68	1.136	122	0.31
		186	0.37			
4.	1.312	143	0.48	1.074	122	0.16
		186	0.25			
5.				1.049	122	0.08
6.				1.021	122	0.01

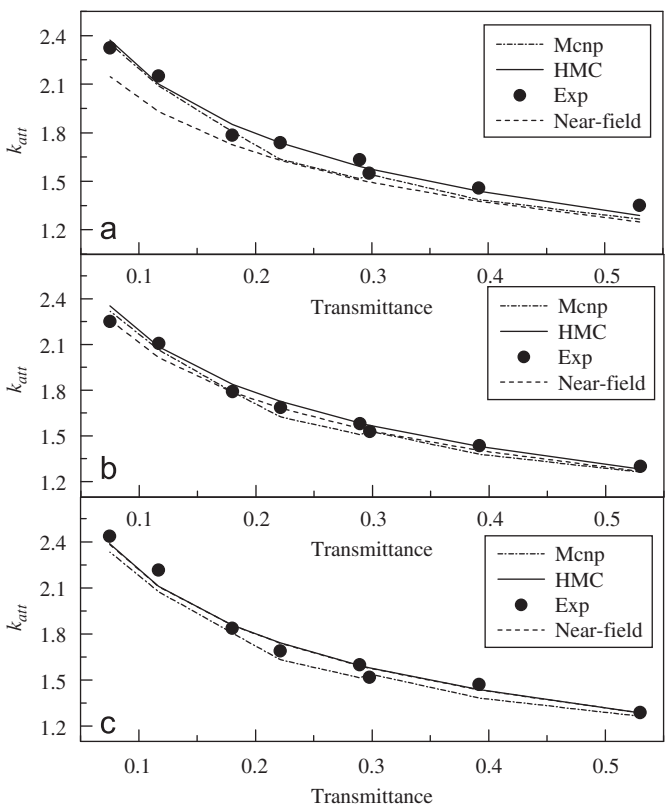

Fig. 5. The attenuation correction factors for cylindrical geometry as a function of transmittance for different sample-to-detector distances (d), (a) $d = 2.5$ cm, (b) $d = 4.5$ cm and (c) $d = 20$ cm.

Table 3
The attenuation correction factors for cylindrical geometry as a function of transmittance at two representative sample-to-detector distances (d).

T	k_{att} at $d = 2.5$ cm				k_{att} at $d = 20$ cm			
	Exp	HMC	MCNP	Near-field	Exp	HMC	MCNP	Near-field
0.53	1.35	1.29	1.27	1.25	1.29	1.29	1.26	1.28
0.39	1.46	1.44	1.39	1.38	1.47	1.44	1.38	1.43
0.30	1.55	1.58	1.54	1.50	1.52	1.58	1.54	1.58
0.29	1.63	1.59	1.52	1.51	1.60	1.59	1.52	1.59
0.22	1.74	1.74	1.64	1.63	1.69	1.74	1.63	1.74
0.18	1.78	1.85	1.81	1.73	1.84	1.86	1.81	1.86
0.12	2.15	2.10	2.09	1.93	2.22	2.11	2.07	2.11
0.07	2.32	2.37	2.35	2.15	2.44	2.39	2.33	2.38

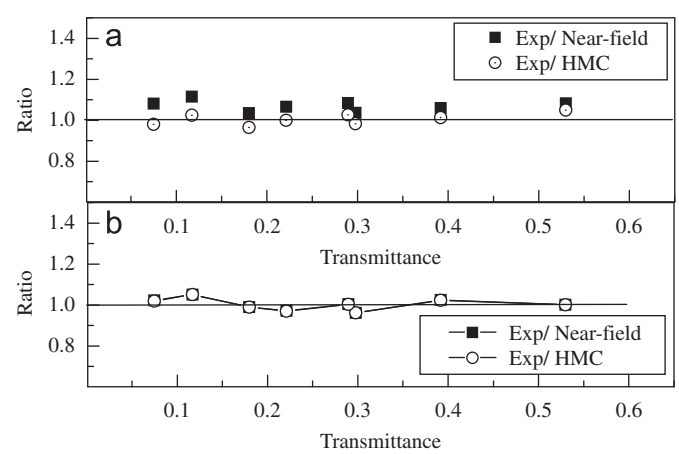

Fig. 6. The ratio of attenuation correction factors for cylindrical geometry as a function of transmittance (a) $d = 2.5$ cm and (b) $d = 20$ cm.

where $\Delta A(m,n)$ is the area element of a circular cross-section of the sample, $L(m,n)$ is the distance of the area element from the point detector, $t(m,n)$ is the distance traveled by the gamma ray in the sample. It is seen from the figure that at $d = 4.5$ and 20 cm, the experimental and the MCNP k_{att} values match well with the HMC k_{att} and near-field formula values. But at $d = 2.5$ cm, the near-field values shows a deviation from the experimental and the MCNP k_{att} values but the HMC approach values match well with the experimental k_{att}. A quantitative comparison can be made from Table 3 which gives the experimental and calculated values of k_{att} at the nearest and the farthest sample-to-detector distance i.e. at $d = 2.5$ and 20 cm. Fig. 6 shows the ratio of experimental attenuation correction factors with the HMC and near-field k_{att} at $d = 2.5$ and 20 cm. At $d = 20$ cm, both the ratios coincide, showing the applicability of all the three approaches at this sample-to-detector distance. A random fluctuation of about 4% seen in the Fig. 6(b) is due to the uncertainty in the experimental k_{att}. However, at $d = 2.5$ cm, the near-field k_{att} deviates from the experimental k_{att} with a deviation reaching to about 10% at lower transmittance while the HMC values are within 4% of the experimental k_{att}'s. This shows that the near-field formula is not valid at very near sample-to-detector distances while the HMC method is valid at all sample-to-detector distances.

Fig. 7 shows the comparison of experimental, HMC and MCNP obtained k_{att} values for disc geometry at three sample-to-detector distances, $d = 2$, 4 and 5 cm. Here also, the theoretical values match reasonably well with the MCNP and experimental k_{att} values at all the distances. This validates the HMC approach over a

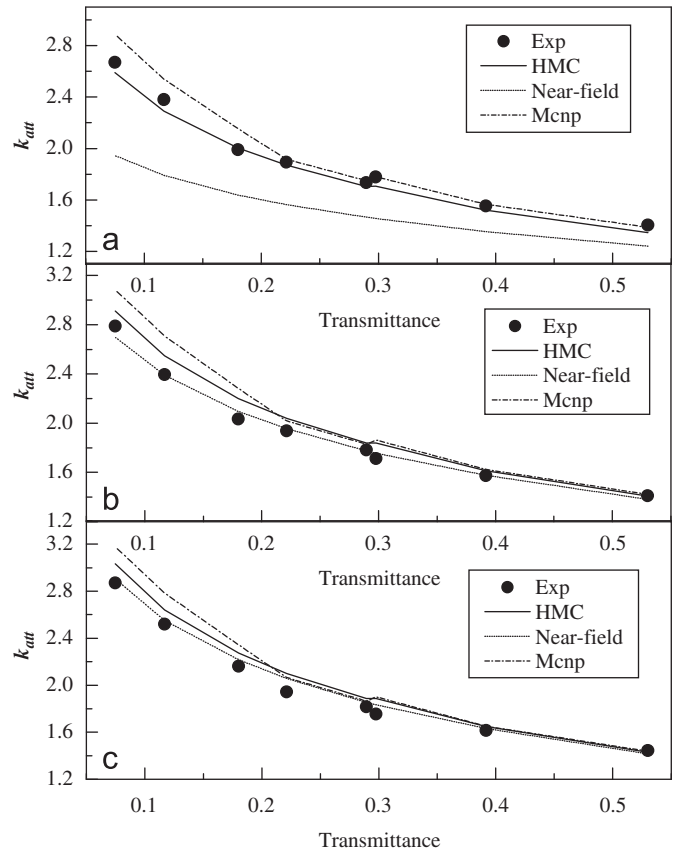

Fig. 7. The attenuation correction factors for disc geometry as a function of transmittance for different sample-to-detector distances (d), (a) $d = 2$ cm, (b) $d = 4$ cm and (c) $d = 5$ cm.

Table 4
The attenuation correction factors for disc geometry as a function of transmittance at two representative sample-to-detector distances (d).

T	k_{att} at $d = 2$ cm				k_{att} at $d = 5$ cm			
	Exp	HMC	MCNP	Near-field	Exp	HMC	MCNP	Near-field
0.53	1.40	1.35	1.39	1.24	1.44	1.43	1.44	1.41
0.39	1.55	1.52	1.57	1.35	1.62	1.65	1.65	1.63
0.30	1.78	1.71	1.78	1.46	1.76	1.89	1.90	1.83
0.29	1.74	1.70	1.75	1.47	1.82	1.89	1.87	1.85
0.22	1.89	1.87	1.92	1.56	1.94	2.10	2.07	2.06
0.18	1.99	2.00	2.15	1.64	2.16	2.27	2.34	2.22
0.12	2.38	2.29	2.54	1.79	2.52	2.64	2.78	2.55
0.07	2.67	2.59	2.89	1.95	2.87	3.03	3.18	2.90

wide range of sample-to-detector distances. Fig. 7 also shows the k_{att} values calculated from a near-field one dimensional model given in [14]

$$k_{att} = [D/d(d+D)]/\sum_{I=1}^{N}\{\exp[-\mu(I-0.5)\Delta x]\}\Delta x/[d+(I-0.5)\Delta x]^2 \tag{7}$$

Here $\Delta x = D/N$ where D is the sample depth, N the number of intervals for the numerical integration and d the distance from the sample surface to the detector surface. The model approximates the sample to be a line sample of depth 'D' and detector to be a point detector. As seen from the figure, the results of the near-field formula deviate significantly at very close sample-to-detector distance i.e. when the sample is almost in the touching configuration with the detector. Table 4 lists the values of

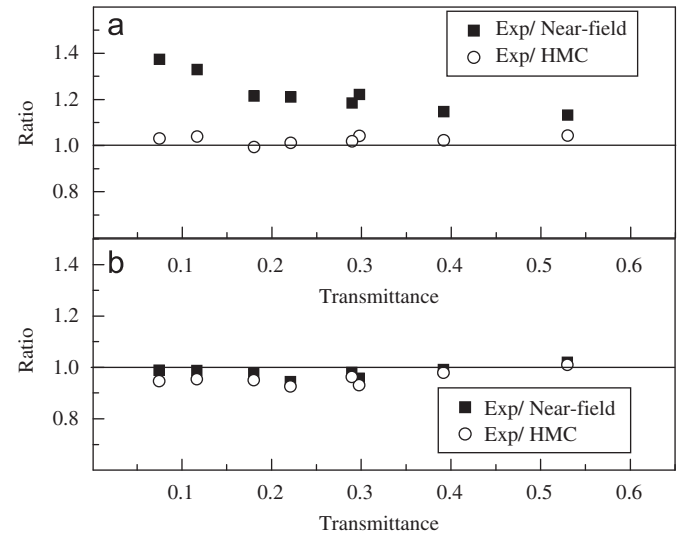

Fig. 8. The ratio of attenuation correction factors for disc geometry as a function of transmittance (a) $d = 2$ cm and (b) $d = 5$ cm.

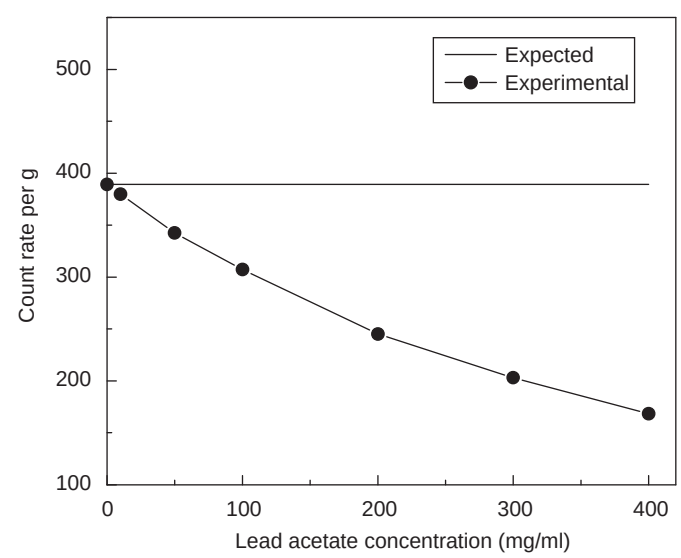

Fig. 9. Count rate at 122 keV per gram of Co-57 activity added as a function of lead acetate concentration. The straight line shows the expected count rate per gram as a function of lead acetate concentration.

experimental and calculated values of k_{att} at $d = 2$ and 5 cm. Fig. 8 shows the ratio of experimental attenuation correction factors with HMC and near-field k_{att} for disc geometry at $d = 2.5$ and 5 cm. It can be seen from the figure that at $d = 5$ cm, both the HMC and near-field k_{att} are within 5% of the experimental k_{att} but at $d = 2.5$ cm, the near-field k_{att} deviates significantly from experimental k_{att} at all transmittances with a deviation of about 35% in lower transmittance range while the HMC k_{att} is within 4% of the experimental k_{att} at all transmittance. This implies that the near-field formula, although said to be valid at close sample-detector geometry, starts working only beyond certain sample-to-detector distance, whereas the HMC approach is found to be applicable at all sample-to-detector distance.

3.2. Box-shaped samples

Fig. 9 shows the experimental count rate per gram of ^{57}Co activity added as a function of lead acetate concentration for the

box samples. The straight line in the figure shows the expected count rate per gram of cobalt activity as a function of lead acetate concentration. Here also, due to gamma ray attenuation, there is a deviation from linearity as lead acetate concentration increases.

The experimental k_{att} for these box-shaped samples were determined by taking the ratio of count rate per gram of the blank solution to the count rate per gram of the sample solution. The HMC calculations were performed for the lead acetate solutions in cubic plastic vials. The dimensions of the samples, the corresponding measured densities and the linear attenuation coefficients are given in Table 2. The diameter of the detector was taken to be 5 cm as given in the manual. These k_{att} were compared with the far-field k_{att} [14] given by

$$k_{att(box)} = \frac{\mu_L x}{[1 - \exp(-\mu_L x)]} \tag{8}$$

Fig. 10 shows the experimental, far-field and HMC method k_{att} for box-shaped sample at sample-to-detector distances, $d = 2.6$, 5.0 and 20.1 cm. At close distance i.e. at $d = 2.6$ and 5.0 cm, the experimental k_{att} deviates significantly from far-field values as expected but matches quite well with the HMC k_{att}. When the sample-to-detector distance is 20.1 cm, all the three attenuation correction factors match well. Table 5 lists the experimental and calculated values of k_{att} at $d = 2.6$ and 20.1 cm. Fig. 11 shows the ratio of experimental k_{att} to the HMC and far-field k_{att}'s at $d = 2.6$ and 20.1 cm. At $d = 20.1$ cm, both the ratios are within 4% but at $d = 2.6$ cm, the deviation between experimental and far-field values is about 14% at lower transmittance while the HMC values matched within 4%. This again shows that the HMC approach is valid at all sample-to-detector distances. Since, no analytical expression for near-field geometry is available, the HMC method can conveniently be used for box-shaped sample at near-field geometry.

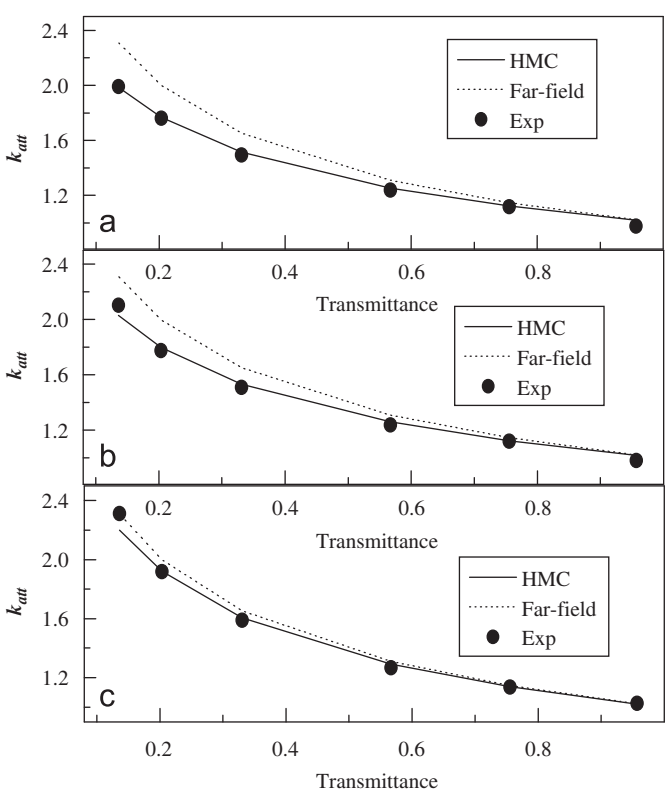

Fig. 10. The attenuation correction factors for box geometry as a function of transmittance for different sample-to-detector distances (d), (a) $d = 2.6$ cm, (b) $d = 5.0$ cm and (c) $d = 20.1$ cm.

Table 5
The attenuation correction factors for box geometry as a function of transmittance at two representative sample-to-detector distances (d).

	k_{att} at $d = 2.6$ cm			k_{att} at $d = 20.1$ cm		
T	Exp	HMC	Far-field	Exp	HMC	Far-field
0.96	0.98	1.02	1.02	1.02	1.02	1.02
0.76	1.12	1.12	1.15	1.14	1.14	1.15
0.57	1.24	1.25	1.31	1.27	1.29	1.31
0.33	1.49	1.51	1.65	1.59	1.60	1.65
0.20	1.76	1.77	2.00	1.92	1.92	2.00
0.14	1.99	1.99	2.31	2.31	2.20	2.31

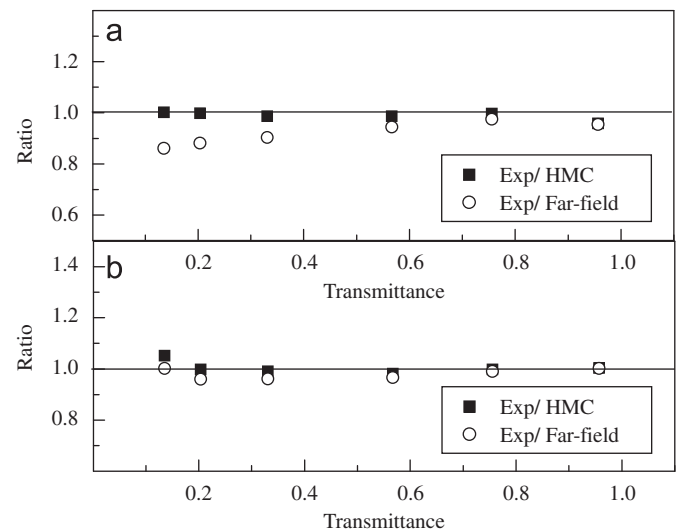

Fig. 11. The ratio of attenuation correction factors for box geometry as a function of transmittance (a) $d = 2.6$ cm and (b) $d = 20.1$ cm.

4. Conclusion

The attenuation correction factors computed from HMC method and Monte Carlo simulation for the cylindrical, disc and box geometry agreed well with the experimental k_{att} values at all sample-to-detector distances. This indicates that the Hybrid Monte Carlo approach can be applied over all sample-to-detector distance. It was observed that the near-field formulas available in literature are applicable only after certain sample-to-detector distance. The advantage of the HMC method is that it is free of any assumptions regarding sample-detector geometry. Also, the HMC approach is common to all sample geometries and sizes.

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